20–STRUCTURAL STRENGTH

1. Introduction to Naval Architecture – Muckle/D.A. Taylor
2. Naval Architecture for Marine Engineers – Reeds
3. Ship Stability for Masters & Mates
4. Maritime Engineering Reference Book

20.1 Loads on Ship Structure

A ship at rest in still water is primarily subjected to the following static loads:

- Cargo weights
- Water pressure

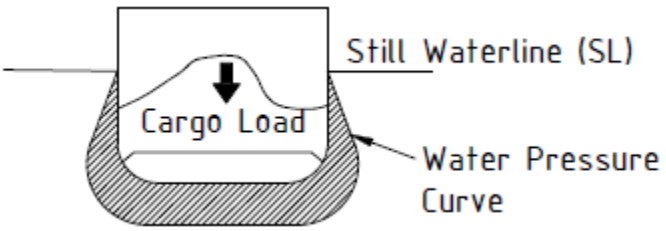


Figure 20.1 – Water Pressure in Still Water

- Load distribution

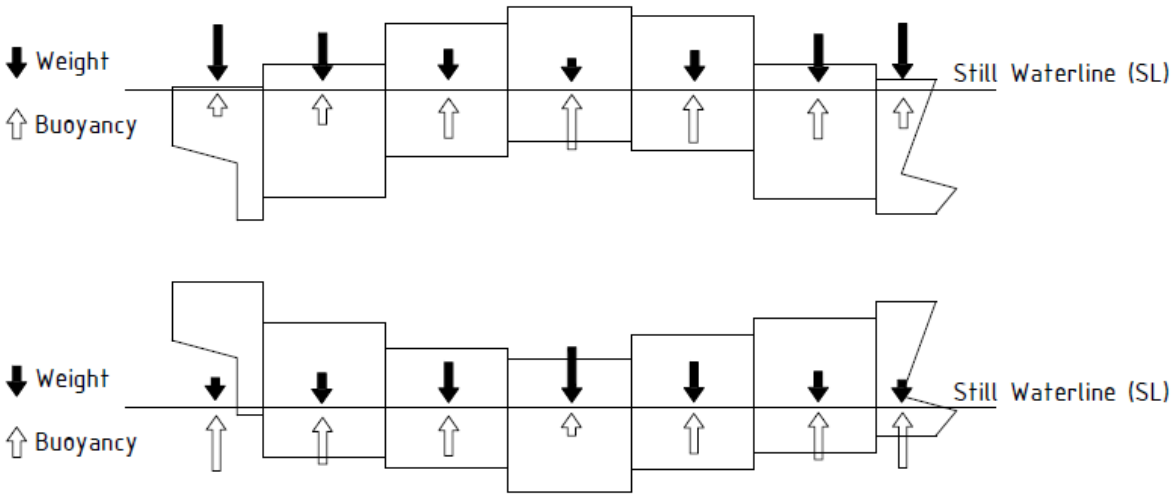


Figure 20.2 – Longitudinal Load Distribution in Still Water

When in motion at sea on the other hand dynamic loads are experienced. These include:

- Acceleration forces due to rolling, pitching, heaving, yawing, surging and swaying motions
- Variable water pressures due to alternate formation of crests and troughs
- Longitudinal bending due to wave actions

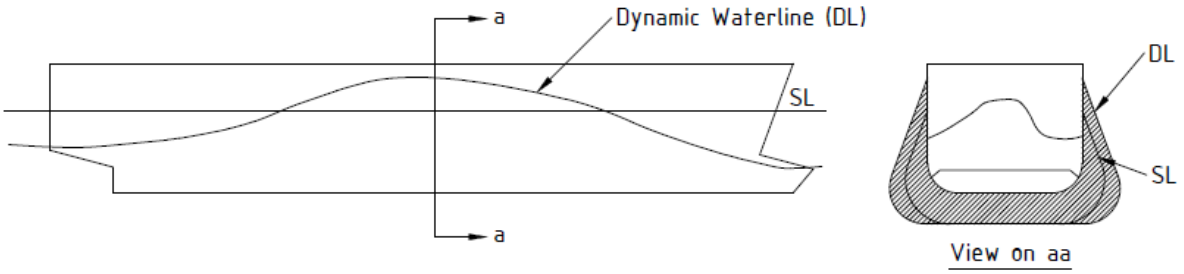


Figure 20.3 – Longitudinal Bending & Water Pressure due to Wave Hogging

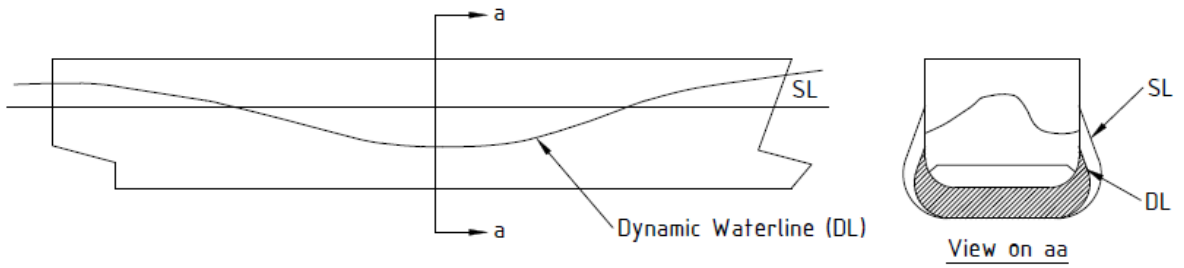


Figure 20.4 – Longitudinal Bending & Water Pressure due to Wave Sagging

The resultant effect in the longitudinal direction is a combination of still water and wave bending which alternates between hogging and sagging as the wave fronts passes. See Figure 20.4.

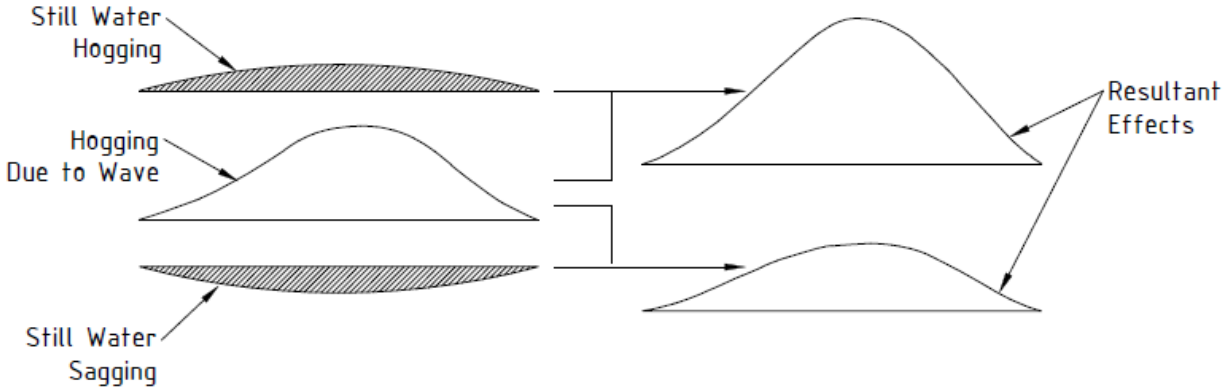


Figure 20.5 – Coupling of Wave Hogging & Still Water Bending

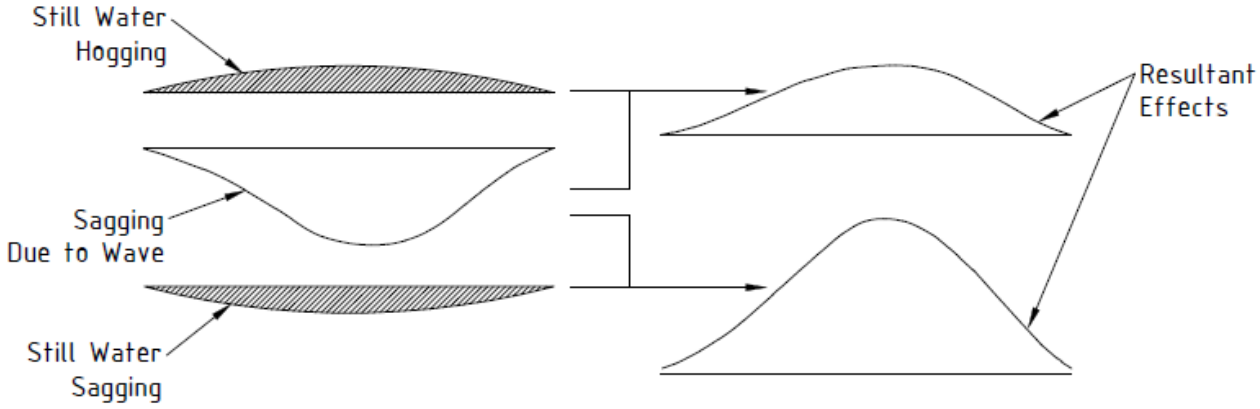


Figure 20.6 – Coupling of Wave Sagging & Still Water Bending

20.2 Shearing Force and Bending Moments

When opposing forces are aligned the resulting effect will be compression. Shearing Forces (SF) are unaligned forces that tend to push one part of a body in one specific direction and the other in the opposite direction.

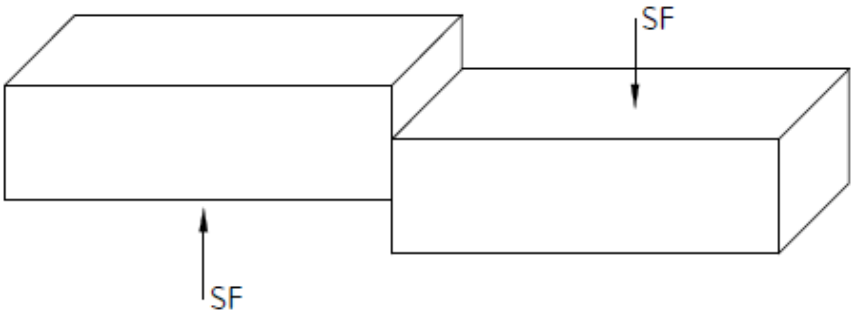


Figure 20.7 – Shearing Forces

Bending Moments (BM) are moments of forces that tend to cause rotation or curvature in the profile of structures.

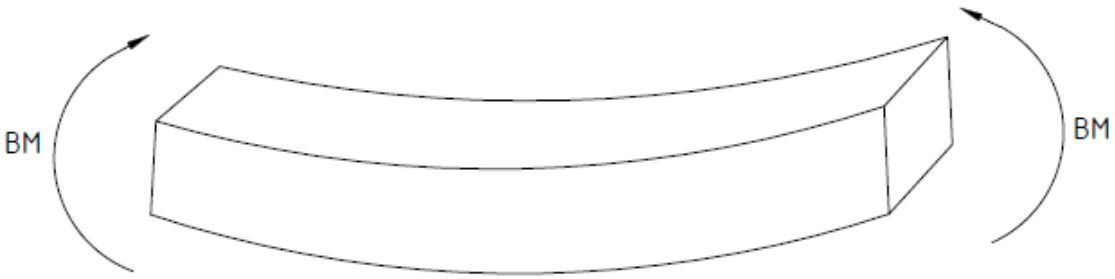


Figure 20.8 – Bending Moments

20.2.1 Still Water Shearing Forces and Bending Moments:

With reference to Figure 20.2, if the mass per unit length is (m) and the immersed cross-sectional area is (A) then in force terms:

$$Buoyancy\ per\ unit\ length = Volume \times Density \times g = A \times 1 \times \rho \times g$$

And hence the net force or:

$$Load\ Distribution/_{Unit\ length} = {}^{Buoyancy}/_{Unit\ length} - {}^{Weight}/_{Unit\ length}$$

$$= (\rho g A - mg)$$

$$SF = \int (\rho g A - mg) dx$$

And

$$BM = \iint (\rho g A - mg) dx \,.\, dx$$

From which it shows that the area of the Load diagram to one side at any point represents the SF at that point, the area of the SF diagram to one side at any point represents the BM at that point

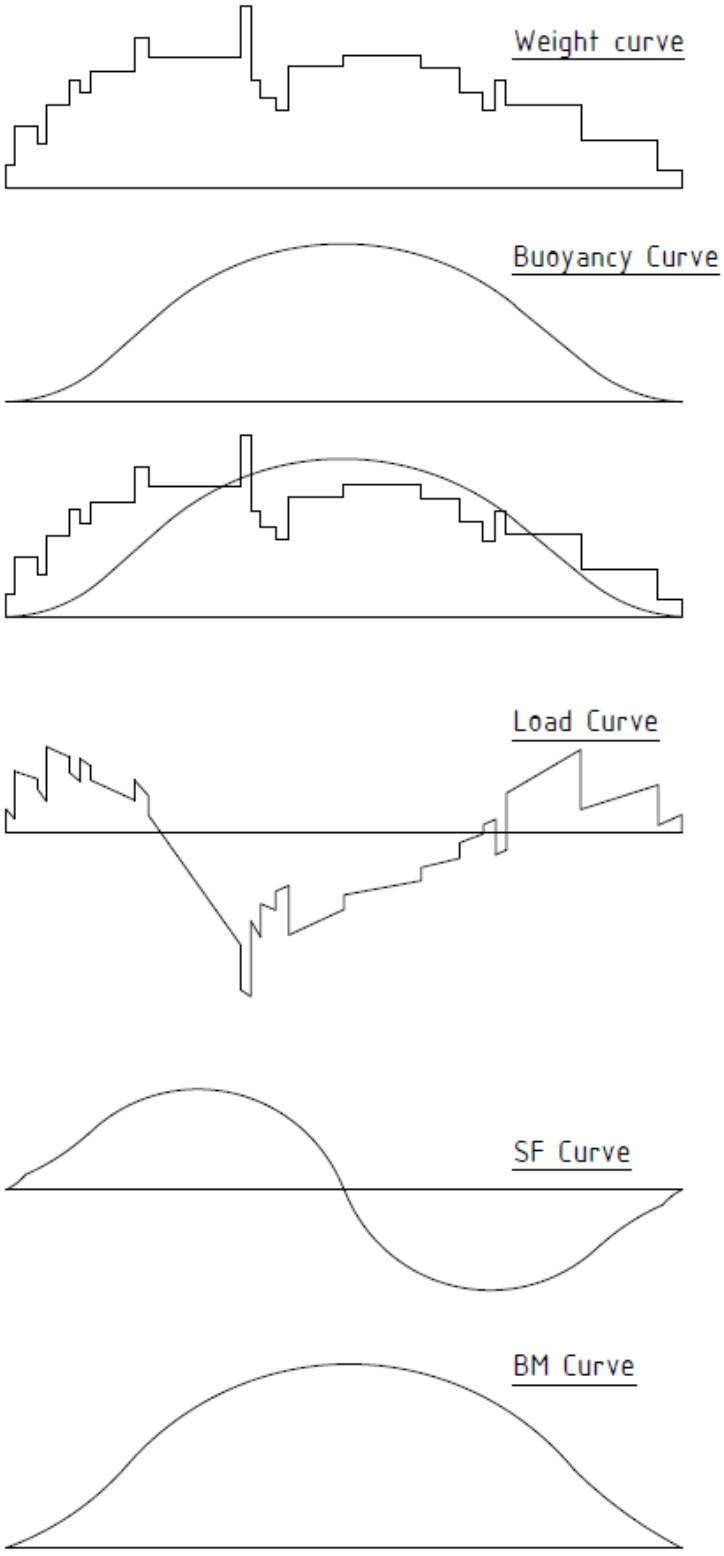
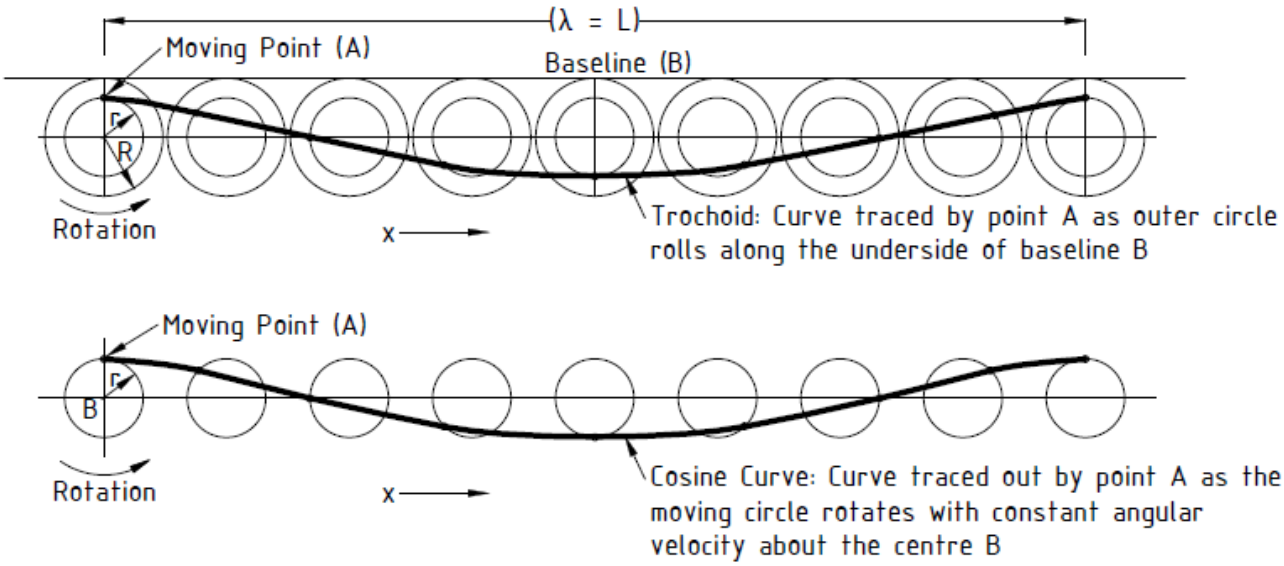


Figure 20.9 – SF & BM Curves

20.2.2 Shearing Forces and Bending Moments in Waves
These are dynamic in nature. However, by reducing the problem to an equivalent static one it is possible to calculate the responses imposed on the hull.

It is generally agreed that the pattern of ocean waves follows the contours of a trochoid, ***a curve traced by a point on a radius of a circle rotating along a straight line***, thereby giving a wave height equal to (0.607√L) at a wave length equal to length of ship (L).



20.10 – Comparison of Trochoid & Cosine Curve

However, by assuming the pattern to be that of a cosine curve instead of a trochoid, the wave height at any point (x) above the still water line level will be given by:

$$h = \left(\frac{0.607\sqrt{L}}{2}\right) cos\left(\frac{2\pi x}{L}\right)$$

From which:

$$Buoyancy/_{metre} = B \times \left(\frac{0.607\sqrt{L}}{2}\right) cos\left(\frac{2\pi x}{L}\right) \rho g$$

$$SF = \int \left(B \times \left(\frac{0.607\sqrt{L}}{2}\right) cos\left(\frac{2\pi x}{L}\right) \rho g\right) dx$$

And:

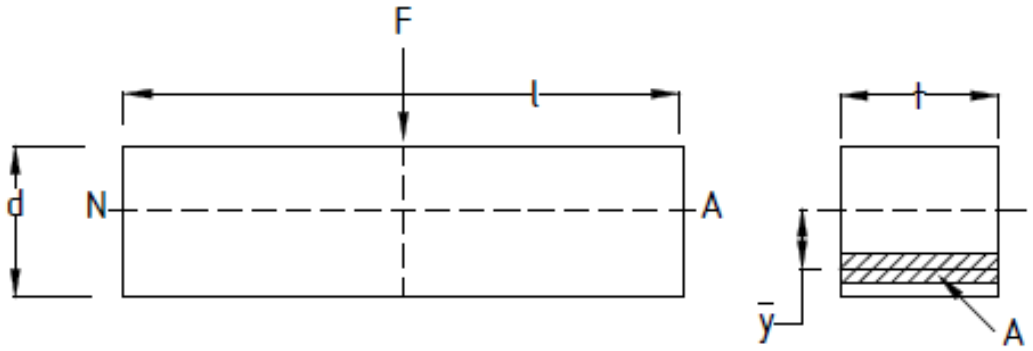
$$BM = \iint \left(B \times \left(\frac{0.607\sqrt{L}}{2}\right) cos\left(\frac{2\pi x}{L}\right) \rho g\right) dx.dx$$

20.3 Shear Stress and Bending Stress

20.3.1 Shear stress:

Shear stress acts tangentially to the hull section and is calculated by:

$$q = \frac{FA\bar{y}}{I_{NA}t}$$



20.11 – Shear Stress

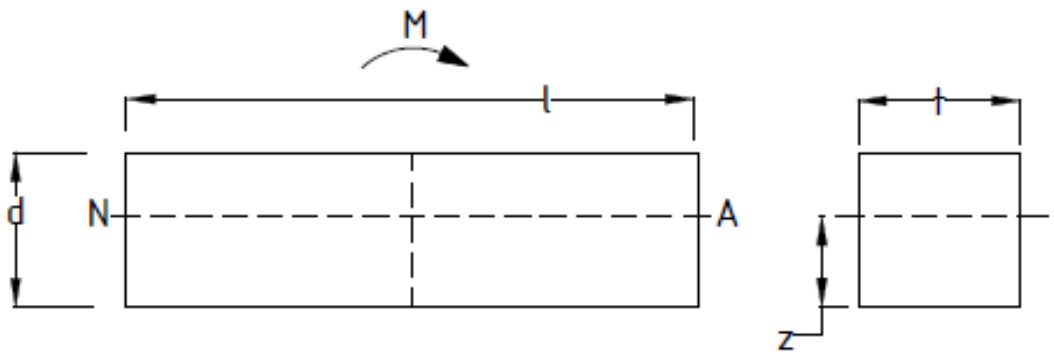
Where:

- F = Shearing force at section
- A = Cross-sectional area of section from the neutral axis of bending
- $\bar{y}$ = The distance of the centroid of the area from the neutral axis
- I_{NA} = Second moment of area of section about the neutral axis
- t = Thickness of section at position y

20.3.2 Bending stress:

Bending due to hogging or sagging puts a ships structure into tension (longitudinal elongation) or compression (shortening longitudinally) referred to as bending stress. Bending stress is represented by (σ) and is determined by:

$$\sigma = \frac{M}{I_{NA}/_{z}}$$



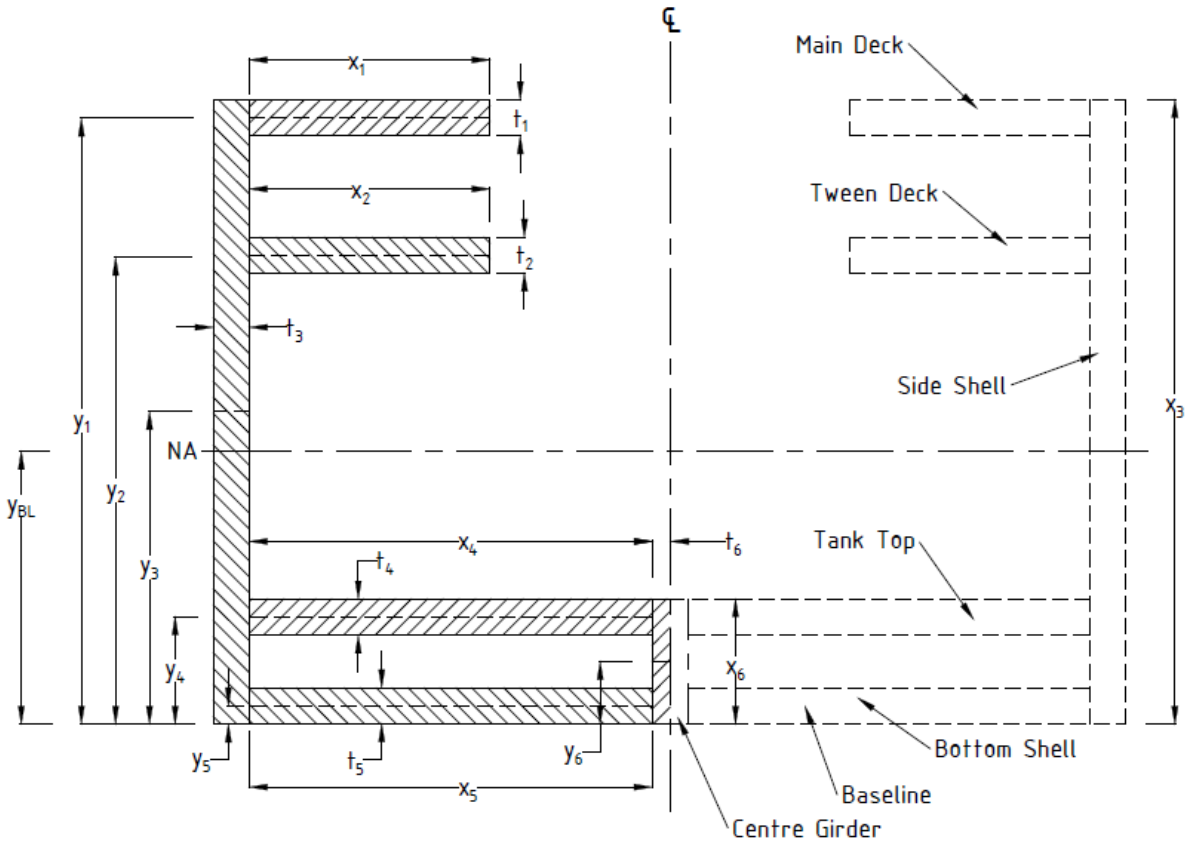
20.12 – Bending Stress

Where:

- M = The bending moment about neutral axis
- z = The distance from extreme edge of section from the neutral axis
- I_{NA} = The second moment of area about the neutral axis
- I_{NA}/z = The 'section modulus'

Bending stress increases from zero at the neutral axis to maximum at the decks and bottom which are farther away.

Calculation of section modulus for a typical half-section of a general cargo ship as illustrated in Figure 20.5 is as follows.



20.13 – Hull Section at Midship

Item	Scantlings (x, t)	I_{na} of Item	Area (a)	Lever (y)	1ST Moment (ay)	2ND Moment (ay^2)
	(m)	(m^4)	(m^2)	(m)	(m^3)	(m^4)
Main Deck	x_1, t_1	$(x_1 * t_1^3)/12$	$x_1 * t_1 = a_1$	y_1	$a_1 * y_1$	$a_1 * y_1^2$
Tween Deck	X_2, t_2	$(x_2 * t_2^3)/12$	$X_2 * t_2 = a_2$	y_2	$a_2 * y_2$	$a_2 * y_2^2$
Side Shell	x_3, t_3	$(t_3 * x_3^3)/12$	$X_3 * t_3 = a_3$	y_3	$a_3 * y_3$	$a_3 * y_3^2$
Tank Top	X_4, t_4	$(x_4 * t_4^3)/12$	$X_4 * t_4 = a_4$	y_4	$a_4 * y_4$	$a_4 * y_4^2$
Bottom Shell	X_5, t_5	$(x_5 * t_5^3)/12$	$X_5 * t_5 = a_5$	y_5	$a_5 * y_5$	$a_5 * y_5^2$
Centre Girder	X_6, t_6	$(t_6 * x_6^3)/12$	$X_6 * t_6 = a_6$	y_6	$a_6 * y_6$	$a_6 * y_6^2$
TOTALS		$\Sigma(I_{na})$	$\Sigma(a)$		$\Sigma(ay)$	$\Sigma(ay^2)$

Neutral axis about baseline (BL)

$$y_{BL} = \frac{\sum(ay)}{\sum(a)}$$

$$I_{BL} = \sum I_{na} + \sum(ay^2)$$

$$I_{NA}\,(Half\ section) = I_{BL} - a(y_{BL})^2$$

$$I_{NA}(Total\ section) = 2 \times I_{NA}(Half\ section)$$

Therefore, at the extreme edges:

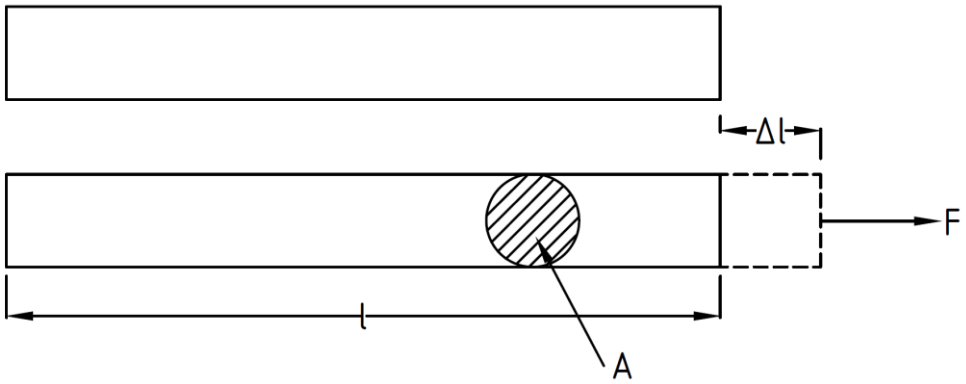
$$Section\ modulus\ for\ base = \frac{I_{NA}(Total\ section)}{z}$$

$$Section\ modulus\ for\ deck = \frac{I_{NA}(Total\ section)}{x_3 - z}$$

20.4 Strain and Hull Deflection

Strain is the effect of stress. It is the relative change or deflection in the dimensions or shape of a body within its elastic limits as a result of applied stress. Strain may be longitudinal or in the form of shear, where:

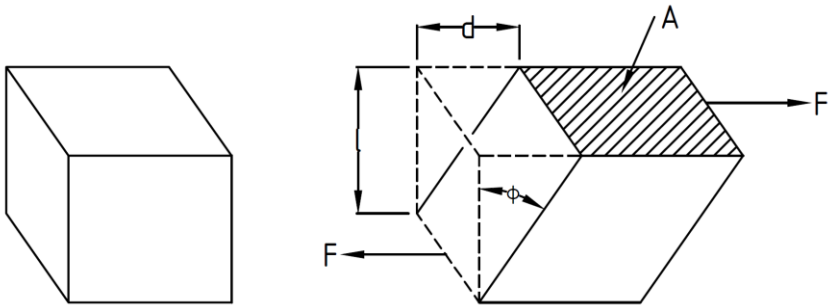
$$Longitudinal\ strain = \frac{Change\ in\ length\ (\Delta l)}{Original\ length(l)}$$



20.14 – Longitudinal Strain

And:

$$Shear\ strain = Angular\ displacement\ (ø)\ of\ body$$



20.15 – Shear Strain

Stress and strain are related by:

$$Modulus\ of\ elasticity = \frac{Stress}{Strain}$$

In the longitudinal direction this is referred to as Young's Modulus:

$$Y = \frac{Longitudinal\ stress}{Longitudinal\ strain} = \frac{F/A}{\Delta l/l} = \frac{Fl}{A.\Delta l}$$

While modulus due to shear stress is referred to as Shear Modulus:

$$S = \frac{Shear\ stress}{Shear\ strain} = \frac{F/A}{\emptyset}$$

Since shear deflection (d) is usually very small, ø approximates to tan ø. Therefore:

$$S = \frac{F/A}{tan\emptyset} = \frac{F/A}{d/l} = \frac{Fl}{d.A}$$

20.5 Loads on Bulkheads & Immersed Planes

20.5.1 When a plate is immersed it is subjected to a load due to liquid pressure. This load (P) is the sum total of all forces that are acting and is considered to be acting perpendicularly at the centrod of the surface which is appropriately referred to as the ***Centre of Pressure***. The load is hence equal to the pressure multiplied by the surface area and is given by:

$$P = \rho g H \times A = \rho g A H$$

Where:

$\rho = Density$
g = Acceleration due to gravity
A = Surface area of plate
H = Distance of the Centre of Pressure from the top edge of the surface

20.5.2 Centres of Pressure of some Typical Plates

Horizontal Plane: When a flat plate is immersed in a liquid with its surface horizontal as in Figure 20.10:

$$h = H$$

And:

$$P = \rho g A h$$

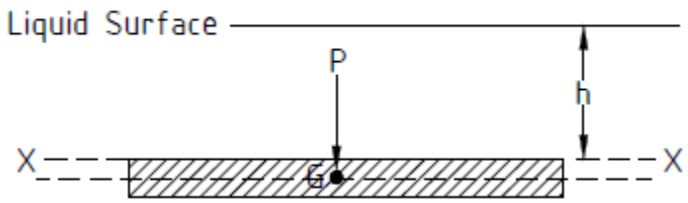


Figure 20.16 – Centre of Pressure on a Horizontal Plate

Inclined Plane:
For an inclined plane, the position of the centre of pressure (H) from the one edge is given by:

$$H = \frac{Second\ Moment\ of\ area\ of\ plate\ section\ about\ the\ edge}{First\ Moment\ of\ area\ of\ plate\ section\ about\ the\ edge}$$

Inclined Rectangular Plane: If an inclined rectangular plate is considered as in Figure 20.11, the position of the centre of pressure from the top edge (XX) of the plate is given by:

$$H = \frac{I_{NA} + A\left(\frac{D}{2}\right)^2}{A\left(\frac{D}{2}\right)} = \frac{\frac{BD^3}{12} + BD\left(\frac{D}{2}\right)^2}{BD\left(\frac{D}{2}\right)} = \mathbf{\frac{2}{3}D}$$

Hence from the liquid surface the position will be:

$$= h + H.sin\theta = h + \frac{2}{3}D \times \sin\theta$$

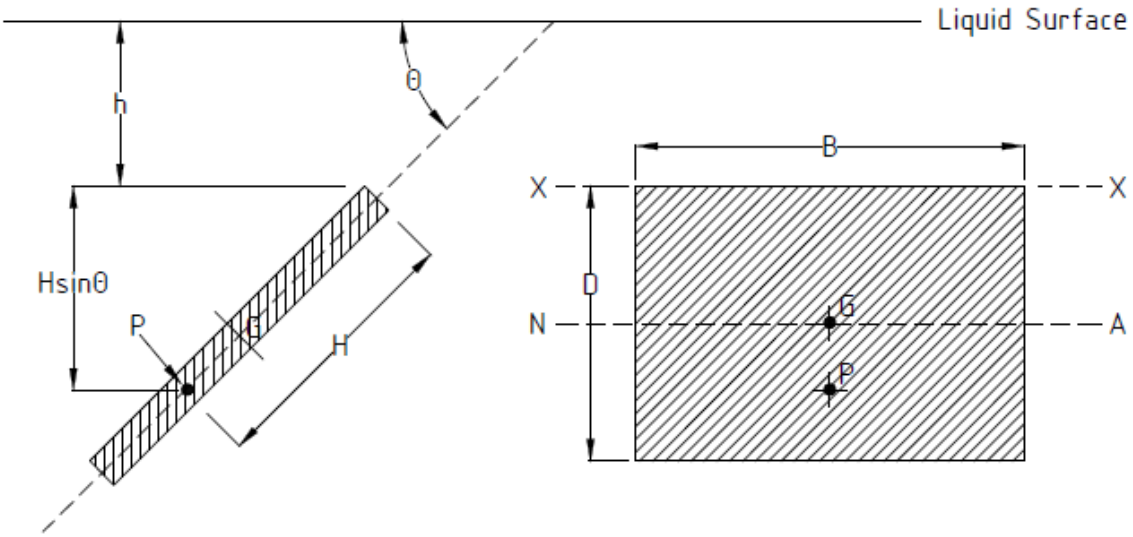


Figure 20.17 – Centre of Pressure on Inclined Horiontal Plate

Inclined Triangular Plane: If an inclined triangular plate is considered as in Figure 20.12, the position of the centre of pressure from the top edge (XX) will be given by:

$$H = \frac{\frac{BD^3}{36} + \frac{BD}{2}\left(\frac{D}{3}\right)^2}{\frac{BD}{2}\left(\frac{D}{3}\right)} = \boldsymbol{\frac{D}{2}}$$

And from the liquid surface the position will be:

$$= h + H.sin\theta = h + \frac{D}{2} \times \sin\theta$$

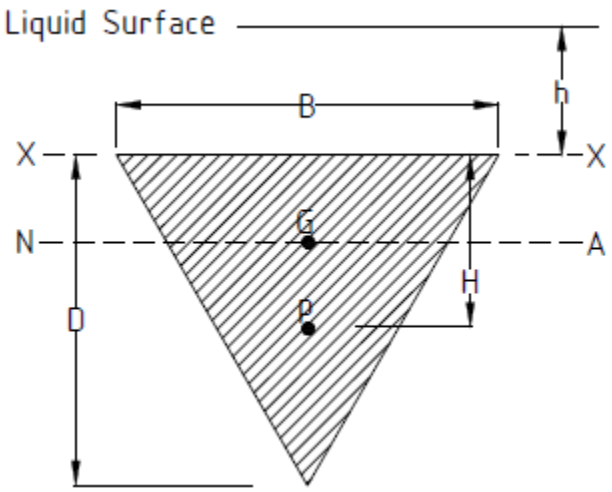


Figure 20.18 – Centre of Pressure on Inclined Triangular Plate

Inclined Circular Plane: For a circular plate the position of the centre of pressure from the top edge (XX) will be given by:

$$H = \frac{\frac{\pi D^4}{64} + \frac{\pi D^2}{4}\left(\frac{D}{2}\right)^2}{\frac{\pi D^2}{4}\left(\frac{D}{2}\right)} = \frac{\mathbf{5}}{\mathbf{8}}\boldsymbol{D}$$

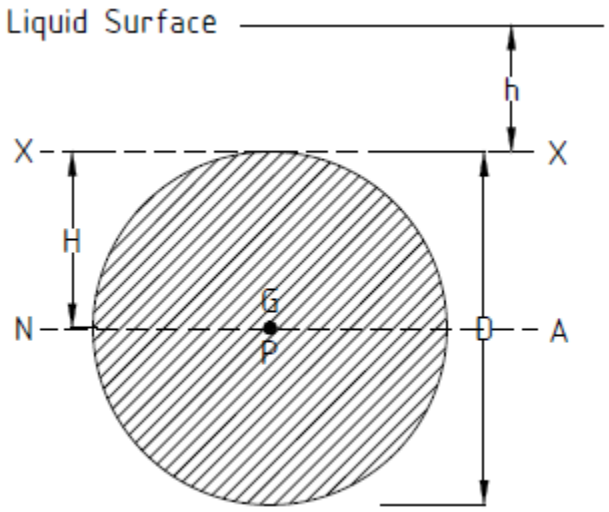


Figure 20.19 – Centre of Pressure on Inclined Circular Plate

And from the liquid surface the position will be:

$$= h + H.sin\theta = h + \frac{5}{8}D \times \sin\theta$$